Maxwell Thomas Asselmeier

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Education

Georgia Institute of Technology — Atlanta, GA
Ph.D. in Robotics (NSF Graduate Research Fellow) — *Aug. 2021 – May 2027*

University of Illinois at Urbana-Champaign — Champaign, IL
B.S. in Mechanical Engineering (Highest Honors) — *Aug. 2017 – May 2021*

Professional Experience

Mitsubishi Electric Research Laboratories (MERL) — Cambridge, MA
Optimization and Robotics Intern — *May 2026 – Oct. 2026*

- **IROS 2026 Humanoid IKEA Assembly Challenge**: Led a 4-person team to 1st place (of 11), deploying large behavior models for furniture assembly on the Unitree G1 and NVIDIA Jetson Thor via TensorRT.
- Integrated vision and tactile sensors into a novel G1-UMI interface to facilitate human demonstration collection.
- Established a teleoperation system with the PICO 4U headset and SONIC whole-body controller to collect 300 demonstrations for dexterous manipulation, used to post-train the GR00T diffusion head through behavior cloning.
- Curated and re-balanced 100 hours of loco-manipulation demonstrations to mitigate bias towards easier subtasks.
- **VLA Failure Detection**: Built an evaluation framework leveraging MuJoCo physics and Isaac Sim visuals to benchmark model performance under torso and hand pose perturbations over 15,000 trials of kitting tasks.
- Extracted dense visual features from CLIP encoders to convert RGB-D and language prompt inputs into affordance heatmaps for zero-shot sim-to-real transfer of learned failure predictors with 91% accuracy.
- Performed distributed model training for VLAs (3B parameters) across 8 NVIDIA A100 GPUs with Slurm.

Institute for Human and Machine Cognition (IHMC) — Pensacola, FL
Humanoid Behaviors Intern — *May 2021 – Aug. 2021*

- **Atlas Perception Engine**: Designed and fabricated a custom multi-modal perception system including LiDARs, cameras, and IMUs; integrated into on-robot software stack for the Boston Dynamics Atlas hydraulic humanoid.
- Integrated visual object tracking algorithms into Atlas's autonomy stack to provide autonomous stair climbing.

Research Experience

Georgia Institute of Technology — Atlanta, GA
Graduate Research Assistant — *Aug. 2021 – Present*

- **Humanoid Skill Chaining**: Develop a world action model trained on trajectory optimization reference data with full-body dynamics constraints to chain 3+ loco-manipulation skills into long-horizon behaviors at 5 Hz.
- Employ sampling-based optimization tools in MuJoCo to generate simulation data for hierarchical skill systems.
- Deploy a whole-body controller trained via RL (PPO) using shaped imitation rewards in Isaac Lab (4096 parallel environments) and PyTorch to track Unitree G1 hand poses on hardware with 3 cm error.
- **QuadPiPS**: Extracted superpixels as mapping features for attention mechanisms in RL-based control.
- Developed a novel panoramic ego-centric image representation in C++ to increase local scene visibility by 200%.
- Trained a tracking controller via imitation learning rewards for terrain traversal including ramps, stairs, and beams.

Publications

5. **M. Asselmeier** and A. Schperberg. Robust Humanoid Foundation Model via Task-Aware Failure Detection and Recovery. Under Review, *IEEE ICRA*. 2027.
4. **M. Asselmeier**, Y. Zhao, and P. A. Vela. QuadPiPS: A Perception-informed Footstep Planner for Quadrupeds With Semantic Affordance Prediction. *IEEE T-RO*. 2026.
3. **M. Asselmeier**, J. Ivanova, Z. Zhou, P. A. Vela, and Y. Zhao. Hierarchical Experience-informed Navigation for Multi-modal Quadrupedal Rebar Grid Traversal. *IEEE ICRA*. 2024.
2. S. Feng, Z. Zhou, J. S. Smith, **M. Asselmeier**, Y. Zhao, and P. A. Vela. GPF-BG: A Hierarchical Vision-Based Planning Framework for Safe Quadrupedal Navigation. *IEEE ICRA*. 2023.
1. B. Mishra, D. Calvert, B. Ortolano, **M. Asselmeier**, L. Fina, S. McCrory, H. E. Sevil, and R. Griffin. Perception Engine Using a Multi-Sensor Head to Enable High-level Humanoid Robot Behaviors. *IEEE ICRA*. 2022.

Skills

Languages	C, C++, Java, MATLAB, Python
Software	CUDA, DDS, Isaac Lab, Isaac Sim, JAX, MuJoCo, PyTorch, ROS, ROS2, Slurm, TensorFlow, TensorRT
Concepts	Behavior Cloning, Diffusion Policies, Imitation Learning, Large Behavior Models, Large-scale Training, Reinforcement Learning, Vision-Language-Action (VLA) Models, World Action Models